SAM-Guided Semantic and Motion Changed Region Mining for Remote Sensing Change Captioning

Futian Wang, Mengqi Wang, Xiao Wang, *Member, IEEE*, Haowen Wang, and Jin Tang

***Abstract*—Remote sensing change captioning (RS-CC) is an emerging and popular research task that aims to describe, in natural language, the content of interest that has changed between two remote sensing images captured at different times. Existing methods typically employ CNNs/Transformers to extract visual representations from the given images or incorporate auxiliary tasks to enhance the final results, with weak region awareness and limited temporal alignment. To address these issues, this article explores the use of the segment anything model (SAM) foundation model to extract region-level representations and inject region-of-interest knowledge into the captioning framework. Specifically, we employ a CNN/Transformer model to extract global-level vision features, leverage the SAM foundation model to delineate semantic- and motion-level change regions, and utilize a specially constructed knowledge graph (KG) to provide information about objects of interest. These heterogeneous sources of information are then fused via cross-attention, and a Transformer decoder is used to generate the final natural language description of the observed changes. Extensive experimental results demonstrate that our method achieves competitive performance across multiple widely used benchmark datasets, especially in terms of BLEU-4 and CIDEr-D scores. The source code of this article has been released on https://github.com/Event-AHU/SAM_ChangeCaptioning**

***Index Terms*—Knowledge-guided learning, remote sensing change captioning (RS-CC), segment anything model (SAM).**

I. Introduction

THE goal of the remote sensing change captioning (RS-CC) task [1] is to express, in natural language, the changes in objects of interest between two given remote sensing images captured at different times. This task can be widely applied in urban planning and land-use monitoring, disaster emergency response, environmental and ecological conservation, as well as military and security surveillance. Although some progress has been made in recent years, significant challenges remain, including difficulties in multitemporal image alignment and registration, the complexity and diversity of change semantics, insufficient fine-grained semantic understanding, and misalignment between language generation and visual changes. Thus, high-quality remote sensing image change captioning (CC) remains an unsolved problem.

Remote sensing CC (RS-CC) has recently emerged as an important task for remote sensing spatiotemporal vision–language understanding. Different from conventional binary or semantic change detection, which mainly focuses on locating changed regions or assigning change categories, RS-CC aims to generate human-readable descriptions that explain what has changed between multitemporal remote sensing images. Existing studies in this field generally follow an encoder–fusion–decoder paradigm. Specifically, bitemporal images are first encoded by visual backbones to extract spatial representations, then cross-temporal feature interaction or fusion modules are employed to identify changed regions, and finally language decoders are used to generate natural language descriptions. Early methods mainly adopted CNN-based visual encoders and RNN-based language decoders [2], [3], [4], establishing the basic pipeline of RS-CC. Subsequently, Transformer-based architectures and attention mechanisms [1], [5], [6], [7] were introduced to enhance global context modeling, multiscale feature interaction, and change-aware representation learning. More recently, prompt-based methods [8], Mamba-based models [9], large language model (LLM)-driven approaches [10], [11], [12], and multitask frameworks jointly learning change detection and CC [13], [14], [15] have further improved the semantic expression ability of RS-CC models.

Despite these advances, existing methods still face several limitations. First, many models mainly rely on implicit feature differencing or generic feature fusion to capture changes, while the discrimination between semantically meaningful changes and pseudo-changes caused by illumination, seasonal variation, shadows, or background noise remains insufficient. This may cause the generated captions to focus on insignificant local variations rather than truly relevant changes. Second, object-level semantic cues and contextual relationships among common remote sensing objects, such as *buildings, roads, vegetation, and bridges*, are still not fully exploited. These cues are essential for generating accurate and human-understandable change descriptions, especially when multiple objects or complex spatial relationships are

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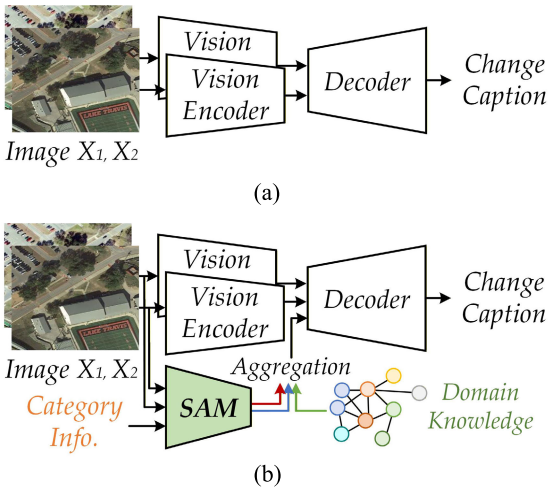


Fig. 1. Comparison between the (a) existing CC framework and (b) our newly proposed SAM-guided semantic and motion changed region mining for RS-CC.

involved. Third, temporal evolution information, such as object appearance, disappearance, expansion, replacement, or structural transformation, is often not explicitly modeled, which limits the ability of existing methods to describe fine-grained changes in a semantically consistent manner. Therefore, a key question naturally arises: "*How can we effectively mine fine-grained change regions while distinguishing semantically relevant changes from insignificant or noisy variations?*"

In this article, as shown in Fig. 1, we propose a novel RS-CC framework that synergistically combines pretrained foundation models, multilevel visual representation, and structured domain knowledge. A key aspect of our framework is to exploit SAM as an external region-aware perceptual prior and further organize its outputs into change-captioning-oriented representations. This is also in line with recent studies highlighting the importance of explicit object-level grounding and region-aware perception in remote sensing understanding [16], [17]. Unlike conventional methods that treat images holistically, we leverage SAM to explicitly delineate regions exhibiting semantic-level and motion-level changes between image pairs. This enables our model to answer not only "*where did the change occur?*" but also "*what changed?*" and crucially, "*is this change relevant or of interest?*," a capability essential for generating accurate and meaningful descriptions. In addition, our framework goes further by incorporating a specially constructed knowledge graph (KG) that encodes prior information about objects commonly involved in meaningful scene changes. This knowledge acts as a semantic prior, guiding the captioning process with contextual cues that pure visual analysis might miss. Recent studies on graph-structured reasoning also suggest that more reliable relation modeling helps improve structured semantic representation [18]. The heterogeneous signals are then effectively fused through a cross-attention mechanism, allowing the model to dynamically weigh visual evidence against semantic expectations. Finally, a Transformer-based decoder is adopted to generate fluent and precise natural language captions. Our work bridges the gap between low-level change detection and high-level semantic interpretation, offering a more robust and interpretable solution to this complex multimodal task. An overview of our proposed framework can be found in Fig. 2.

To sum up, the main contributions of this article can be summarized as follows.

1) We propose a novel RS-CC framework that leverages the segment anything model (SAM) to explicitly identify semantic- and motion-level change regions, enabling accurate localization of what changed and whether it matters.

2) We construct a semantic KG using LLMs and integrate it into the captioning pipeline as a contextual prior. This design strengthens the semantic association between visual changes and language generation, thereby improving the coherence, relevance, and interpretability of the generated descriptions.

3) Extensive experiments on three widely used benchmark datasets demonstrate the effectiveness and competitiveness of the proposed approach.

II. Related Works

A. Remote Sensing CC

RS-CC is one of the core tasks in the field of vision–language understanding for remote sensing time-series images. Jhamtani and Berg-Kirkpatrick [2] proposed the CC task and introduced a new dataset along with a Siamese CNN–RNN architecture to capture significant differences between similar images. Subsequently, two small-scale RSICC datasets, namely LEVIR CCD and Dubai CCD, were released, focusing on RS-CC. To further advance the research, Liu et al. [1] provided a new large-scale RSICC dataset, LEVIR-CC, and built an RSICC network based on the Transformer architecture, effectively demonstrating the adaptability of Transformer structures in RSICC tasks. Chang and Ghamisi [5] integrated a Siamese CNN with an attention-based Transformer encoder–decoder to dynamically locate change regions. Cai et al. [6] introduced ICT-Net, employing cross-gated attention and adaptive fusion to enhance change representation. Sun et al. [19] proposed a sparse-focused Transformer (SFT) that reduces computational cost while maintaining competitive performance. Zhou et al. [7] developed a single-stream feature extraction network (SEN) with contrastive pretraining and cross-attention modules to improve efficiency. Zhu et al. [14] introduced Change3D, which treats bi-temporal images as video frames to achieve unified modeling of change detection and captioning. Sun et al. [20] introduced a diffusion-based data distribution learning framework with frequency-domain noise filtering to overcome overfitting and improve model generalization. More recently, Yang et al. [12] proposed KCFI, which enhances RS-CC by integrating key change feature perception, pixel-level change detection, and an instruction-tuned large language decoder. Li et al. [21] proposed difference semantic prior guidance (DSPG), which enhances fine-grained change modeling through dual-branch difference encoding and semantic-prior-guided learning. León et al. [22] introduced MVLT-LoRA-CC, which improves RS-CC through multimodal fusion of ViT, LLM, and selective LoRA adaptation. Tuzlupinar et al. [23] proposed X-Change, an RGB-inspired spectral-aware framework that combines rule-driven spectral-index supervision with

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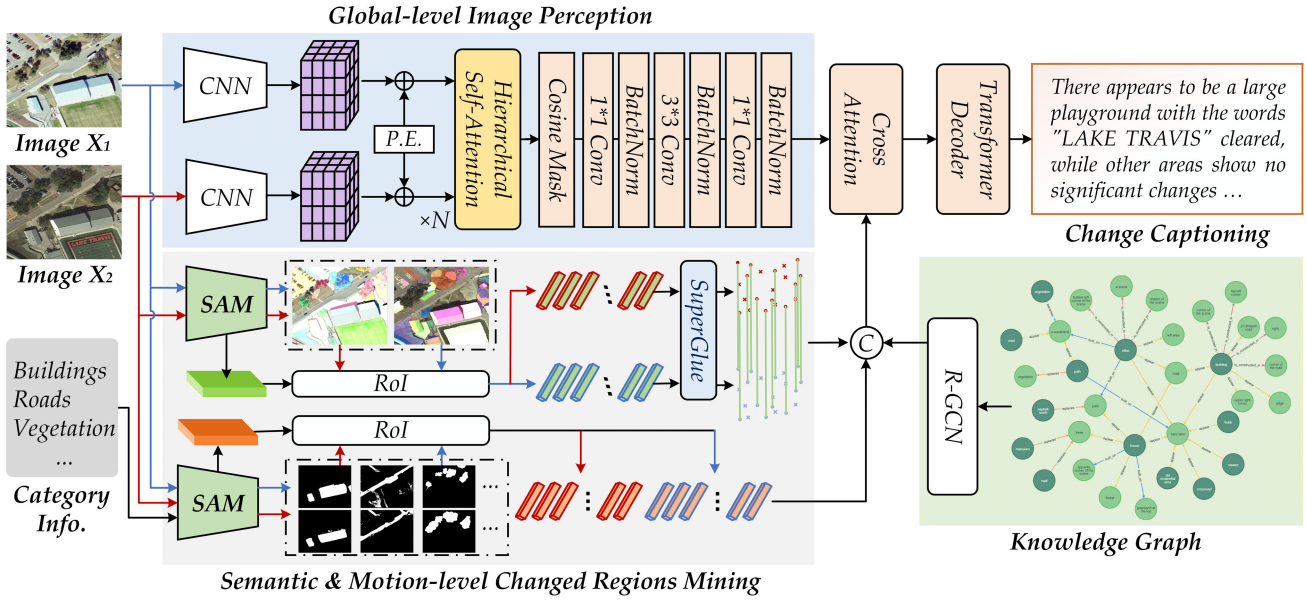


Fig. 2. Illustration of our proposed RS-CC by SAM-guided semantic, motion changed region mining and KG.

joint segmentation-caption learning, enabling explainable CC from RGB bi-temporal images with performance comparable to multispectral settings. Xue et al. [24] further presented ChangeVG, which unifies CC, classification, counting, and localization within a comprehensive remote sensing change understanding framework based on ChangeIMTI and a dual-granularity enhanced vision–language model. Despite these advances, most existing RS-CC methods still rely primarily on global feature interaction, implicit change modeling, word-level semantic prompting, or heavy pretrained vision–language adaptation, while insufficiently exploring explicit grounding of semantically meaningful changed regions and structured semantic relation modeling among changed objects. As a result, they still face limitations in subtle change perception, fine-grained semantic alignment, and description consistency in complex remote sensing scenes. Different from these works, our method combines SAM-guided changed-region localization with knowledge-graph-guided semantic reasoning to explicitly identify semantically meaningful changed regions and integrate structured semantic priors, thereby improving both the accuracy and semantic consistency of remote sensing change descriptions.

B. Segment Anything Model

The SAM [25] is a large-scale pretrained model with strong zero-shot generalization. Inspired by the prompt strategy in NLP, the SAM segments diverse targets based on various prompt types, including points, bounding boxes, masks, and text. Following SAM's release, numerous studies propose improvements and domain-specific adaptations to enhance its performance and generalization [26], [27], [28], [29], [30].

Cheng et al. [31] proposed SAM-Med2D, which adapts the SAM for medical image segmentation through large-scale fine-tuning and diverse prompting strategies. Chen et al. [32] proposed RSPrompter, combining the SAM with semantic category information to automatically generate prompts for semantic instance segmentation in remote sensing images. Grounded SAM [33] integrates Grounding DINO [34] with the SAM to improve open-vocabulary image segmentation and extend its applicability to more controllable tasks. Recent studies have further explored the integration of the SAM into remote sensing change analysis. For example, Gao et al. [35] showed that the SAM can still be effectively adapted to remote sensing change detection even under limited data conditions, while Zhang et al. [36] demonstrated that incorporating feature interaction with the SAM helps enhance change-aware representation learning. Although existing SAM-based studies demonstrate strong region perception and segmentation capability, they mainly focus on segmentation quality, domain adaptation, or change detection, rather than explicitly modeling semantically meaningful changed regions for RS-CC. This leaves a gap between general-purpose region perception and fine-grained change description in bi-temporal remote sensing scenes. Inspired by these works, we adopt the SAM in this article to analyze remote sensing images and better locate meaningful changed regions for CC.

C. Knowledge Graph

KGs, first introduced by Google in 2012, represent entities and relations through graph structures for knowledge organization and reasoning. With the rise of deep learning and NLP, neural models such as the BiLSTM [37], CNN [38], and Transformer [39] have become dominant in entity and relation extraction. Recently, LLMs have opened new directions for KG construction and application by efficiently extracting, completing, and reasoning over knowledge. The integration of LLMs with KGs, either by using KGs as external memory or by leveraging LLMs to expand and refine them, becomes a major research trend. Several frameworks explore different forms of KG–LLM integration. Kim et al. [40] proposed KG-GPT, combining sentence segmentation, graph retrieval, and inference for fact verification and question answering.

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Shu et al. [41] introduced KG-LLM, converting KG structures into natural language prompts for multihop link prediction. Mo et al. [42] presented KGGen, which improves extracted graphs through iterative clustering and relation grouping.

Graph neural networks (GNNs), such as the R-GCN [43] and CompGCN [44], effectively combine structural and semantic information for KG representation learning and reasoning. Recent research integrates KGs with natural language generation (NLG) to improve factuality and interpretability in descriptive and report generation. Soman et al. [45] proposed KG-RAG for biomedical text generation; Wu et al. [46] introduced a multimodal KG-driven diagnostic text generator; and Wang et al. [47] proposed R2GenKG, which builds a large-scale multimodal medical KG (M3KG) using GPT-4o and R-GCN to generate high-quality X-ray reports. However, most existing KG-based or KG-enhanced methods are primarily developed for general NLP tasks, reasoning, or report generation, and focus on static entity–relation modeling. They have not been sufficiently explored for capturing dynamic change patterns, modeling interactions among changed objects, or providing structured semantic priors tailored to RS-CC. As a result, directly applying these methods to RS-CC may fail to fully bridge the gap between visual change perception and high-level semantic description. Different from these works, we use a relation-aware KG to bridge visual and semantic differences in dual-time remote sensing images, enhancing temporal reasoning and caption generation.

III. Our Proposed Approach

A. Overview

As illustrated in Fig. 2, we propose segment-assisted graph-enhanced change captioner (SAGE-CC), a framework for RS-CC from bi-temporal image pairs acquired at two epochs. Specifically, the bi-temporal scene consistency encoder (Section III-B) extracts shared multiscale features and distills explicit cross-time consistency priors, offering robust global context to support subsequent modules. The SAM-guided change region mining (Section III-C) leverages the SAM to jointly perform motion-level change localization (MCL) of pixel-accurate changed regions and prompt-guided aggregation of semantics into category-aware region representations. Moreover, the remote sensing change graph reasoner (CGR) (Section III-D) constructs a compact KG from training captions and employs relation-aware message passing to generate domain-specific priors on key entities and typical change patterns. Finally, the change-aware language generator (Section III-E) integrates global scene priors, localized region representations, and KG priors, producing concise and faithful change descriptions that clearly articulate *where*, *what*, and *how* the scene changes. Overall, the proposed framework follows a progressive design that jointly models remote sensing changes from the perspectives of entities and relations. SAM-guided region mining provides visually grounded entity-level candidates, while superglue (SG)-based bi-temporal region matching captures relation-level changes among these entities, such as appearance, disappearance, and transformation. Based on these entity- and relation-level cues, the graph reasoner further organizes them into structured semantic priors, thereby connecting motion-level change grounding with high-level semantic reasoning. Training objectives and optimization details are provided in Section III-E.

B. Bi-Temporal Scene Consistency Encoder

This module generates a global scene embedding and explicit pixel-wise consistency and change priors from a bi-temporal image pair.

1) Spatial Feature Backbone: Given a bi-temporal image pair $X = (X_1, X_2)$, a Siamese ResNet-101 [48] with shared parameters extracts feature maps $F_1, F_2 \in \mathbb{R}^{h\times w\times C}$. To preserve spatial structure, a 2-D positional embedding $P \in \mathbb{R}^{h\times w\times C}$ is added to each stream

$$F_i^{(0)} = F_i + P, \quad i \in \{1, 2\} \tag{1}$$

producing coordinate-aware features in a unified representation space.

2) Cross-Time Consistency Modeling: To enhance temporal reasoning, we employ N lightweight attention blocks that jointly refine intratemporal semantics and highlight intertemporal discrepancies. In each block l, every stream first applies multihead self-attention (SA) and then cross-attends (CAs) to the concatenated bi-temporal context, where $H_i^{(l)}$ can be viewed as the query feature and $[H_1^{(l)}, H_2^{(l)}]$ provide the joint contextual information for the interaction. The attention modules in the two temporal branches share parameters

$$H_i^{(l)} = \mathrm{SA}\left(F_i^{(l-1)}\right) + F_i^{(l-1)} \tag{2}$$

$$F_i^{(l)} = \mathrm{CA}\left(H_i^{(l)}, \left[H_1^{(l)}, H_2^{(l)}\right]\right) + H_i^{(l)}, \quad i \in \{1, 2\}. \tag{3}$$

After N blocks, we obtain temporally aligned features $F_1^{(N)}, F_2^{(N)} \in \mathbb{R}^{h\times w\times C}$ and compute a pixel-wise cosine similarity

$$S(u, v) = \frac{\left\langle F_1^{(N)}(u, v, :),\ F_2^{(N)}(u, v, :)\right\rangle}{\left\|F_1^{(N)}(u, v, :)\right\|_2 \left\|F_2^{(N)}(u, v, :)\right\|_2}. \tag{4}$$

The similarity map S is normalized to obtain a *consistency prior* $C_{\mathrm{map}} = (S + 1)/2 \in [0, 1]^{h\times w}$, and its complement is defined as the *change prior* $M_{\mathrm{map}} = 1 - C_{\mathrm{map}}$. We then fuse features under the guidance of these priors

$$G = \mathrm{concat}\Big(F_1^{(N)} \odot C_{\mathrm{map}},\ F_2^{(N)} \odot C_{\mathrm{map}},\ |F_1^{(N)} - F_2^{(N)}| \odot M_{\mathrm{map}},\ M_{\mathrm{map}}\Big) \tag{5}$$

which is processed by a shallow Conv–BN–ReLU head followed by a 1×1 projection to yield

$$E_{\mathrm{img}} = \phi_\theta(G), \quad E_{\mathrm{img}} \in \mathbb{R}^{h\times w\times C_e}. \tag{6}$$

The resulting global embedding E_{img} and priors $\{C_{\mathrm{map}}, M_{\mathrm{map}}\}$ are forwarded to subsequent modules for downstream reasoning.

C. SAM-Guided Change Region Mining

SAM-guided change region mining leverages the SAM [25] to localize pixel-accurate, motion-level change regions and to aggregate semantic-level change features into category-aware region representations from the bi-temporal pair.

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1) Motion-Level Change Localization: Given each image $X_i (i \in \{1,2\})$, the SAM first generates a set of instance masks $\mathcal{M}_i = \{m_{i,n}\}_{n=1}^{N_i}$ and exposes an intermediate dense feature map $S_i \in \mathbb{R}^{h\times w\times C_s}$ from its frozen image encoder. To select reliable and complete region proposals, we filter candidate masks based on confidence scores, region area, nonmaximum suppression, and a minimum-area threshold.

Each mask is converted into a bounding box $r_{i,n}$, and corresponding region features are extracted using RoIAlign

$$\mathcal{X}_i = \text{RoIAlign}\left(S_i, \{r_{i,n}\}\right) \in \mathbb{R}^{N_i\times h_r\times w_r\times C_s}. \tag{7}$$

These region features undergo adaptive average pooling to a fixed spatial size (7×7) followed by a lightweight convolutional head g_θ, yielding compact descriptors

$$\mathcal{Z}_i = g_\theta\left(\text{AAPool}_{7\times 7}(\mathcal{X}_i)\right) \in \mathbb{R}^{N_i\times d} \tag{8}$$

with centroids denoted as $K_i = \{k_{i,n} \in \mathbb{R}^2\}$.

We employ SG [49] with frozen parameters to compute cross-epoch correspondence scores between region descriptors

$$\left[Z_{pq}\right]_{N_1\times N_2} = \text{Superglue}\ (K_1, \mathcal{Z}_1;\ K_2, \mathcal{Z}_2) \tag{9}$$

where each entry Z_{pq} represents the matching confidence between region p from epoch 1 and region q from epoch 2. Higher values indicate stronger geometric and semantic consistency. Reliable matches are determined by applying a threshold τ

$$Z_{pq} = \begin{cases} 1, & \text{if } Z_{pq} > \tau \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

Unmatched instances, potentially corresponding to changed or newly appeared/disappeared regions, are identified by

$$U_1 = \left\{p \,\middle|\, \sum_q Z_{pq} = 0\right\}, \quad U_2 = \left\{q \,\middle|\, \sum_p Z_{pq} = 0\right\}.$$

To incorporate spatial information, centroids of unmatched regions are embedded via positional encoding and concatenated with their respective region descriptors. The final motion-level change representation is obtained by concatenating these enriched features within each image, capturing structural changes at pixel-level precision without assuming object motion.

2) Semantic-Level Change Aggregation: To overcome the limitations of purely visual or manually defined prompts, we introduce Grounding DINO [34] as a semantic grounding method before automatically generating prompt-conditioned bounding boxes for both images X_1 and X_2. These boxes correspond to high-level semantic categories such as *building*, *road*, and *vegetation*. Subsequently, the SAM refines these bounding boxes into detailed segmentation masks, resulting in region-specific dense feature maps $S_i \in \mathbb{R}^{h\times w\times C_s}$.

Given textual prompts $\mathcal{T} = \{t_k\}_{k=1}^K$, we compute their text embeddings $E_T = \text{enc}_T(\mathcal{T}) \in \mathbb{R}^{K\times d}$. For each segmented region, we extract region embeddings $\mathcal{E} \in \mathbb{R}^{J\times d}$ using the same RoI pipeline described above. Image–text similarity is computed via normalized dot products

$$S = \text{Norm}(\mathcal{E})\ \text{Norm}(E_T)^\top \in \mathbb{R}^{J\times K}. \tag{11}$$

Regions are ranked based on their highest similarity scores across all prompts, and the top-q regions are selected

$$\mathcal{I}_{\text{sel}} = \text{Top-}q\left(\max_k S_{:,k}\right). \tag{12}$$

The final semantic-level change representation is constructed by concatenating the embeddings of selected regions from each prompt category. This semantic-aware representation effectively complements the motion-level change features, yielding a more coherent and interpretable representation of change across epochs.

D. Remote Sensing CGR

This module constructs a compact remote sensing change KG from caption corpora and employs relation-aware message passing to derive semantic priors for subsequent reasoning and caption generation tasks.

1) Graph Construction From Captions: Given a set of descriptive captions $\mathcal{C}$ from multitemporal remote sensing datasets, each caption describes objects, land-cover types, and their transformations over time. We apply an LLM-based [50] relation extraction approach specifically adapted to remote sensing change descriptions to extract structured semantic triples $\mathcal{T} \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$, where $\mathcal{E}$ and $\mathcal{R}$ denote the raw entity and relation vocabularies. The extraction process is further guided by task-specific designs, including change-aware relation modeling and entity normalization, to ensure semantic consistency and effectively capture temporal transitions.

For example, for the caption "a crossroad and several buildings appear on the bareland," the LLM-based relation extraction process can extract triples such as $\langle$`crossroad`, `appear-on`, `bareland`$\rangle$ and $\langle$`building`, `appear-on`, `bareland`$\rangle$. These triples explicitly characterize the temporal change relation between changed objects and land-cover categories.

For the entity normalization step, we perform iterative clustering with instruction set $\mathbf{I}_E$ to merge semantically similar entities

$$\mathcal{E}' = \text{Merge}_{\mathbf{I}_E}(\mathcal{E}), \quad \pi_E : \mathcal{E} \to \mathcal{E}'.$$

Similarly, the relation set is normalized to obtain a canonical relation vocabulary $\mathcal{R}'$ with mapping $\pi_R : \mathcal{R} \to \mathcal{R}'$.

All triples are subsequently remapped via π_E and π_R and deduplicated to produce the normalized KG

$$\mathcal{G} = (\mathcal{E}', \mathcal{R}', \mathcal{T}').$$

For computational efficiency, we encode the graph $\mathcal{G}$ using two matrices: the connectivity matrix

$$A_{\text{conn}} = \begin{bmatrix} \text{src}_1 & \text{src}_2 & \dots & \text{src}_m \\ \text{tgt}_1 & \text{tgt}_2 & \dots & \text{tgt}_m \end{bmatrix}$$
$$A_{\text{type}} = \begin{bmatrix} r_1 & r_2 & \dots & r_m \end{bmatrix} \tag{13}$$

where $m = |\mathcal{T}'|$ and $\text{src}_j, \text{tgt}_j \in \{1,\dots,|\mathcal{E}'|\}$ denote indices of head and tail entities, respectively, and $r_j \in \{1,\dots,|\mathcal{R}'|\}$ encodes relation types corresponding to land-use transitions and spatial interactions.

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2) Relation-Aware Graph Encoding: Entities are initialized using contextualized text embeddings. Let $e'_i \in \mathcal{E}'$ denote the ith entity; a pretrained text encoder (e.g., BERT) generates initial node embeddings

$$V = \text{BERT}\left(\left[e'_1, \ldots, e'_{|\mathcal{E}'|}\right]\right) \in \mathbb{R}^{|\mathcal{E}'| \times d_b} \tag{14}$$

with $h_i^{(0)} = V_i$.

We employ a multilayer relational graph convolutional network (R-GCN) [43] to propagate and aggregate relation-specific features across graph nodes

$$h_i^{(l+1)} = \sigma\left(W_0^{(l)} h_i^{(l)} + \sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{N}_i^r} \frac{1}{c_{i,r}} W_r^{(l)} h_j^{(l)}\right) \tag{15}$$

where $\mathcal{N}_i^r$ denotes neighboring nodes of node i linked by relation type r (retrievable from A_{conn} and A_{type}), $c_{i,r}$ is a normalization constant, $W_r^{(l)}$ and $W_0^{(l)}$ are relation-specific and self-loop weights, respectively, and σ is an activation function.

Unlike standard GCN architectures, the R-GCN explicitly conditions message-passing transformations on relation type and edge directionality, enabling it to encode asymmetric dependencies. This is particularly crucial for remote sensing tasks where relationships like "`bareland` → `building`" differ fundamentally from "`building` → `bareland`" in semantic meaning and temporal direction.

Instead of compressing the graph into a single graph-level vector, we retain the final node representations as relation-aware graph memory tokens. Specifically, let

$$F_{\text{kg}} = W_g H^{(L)} \in \mathbb{R}^{|\mathcal{E}'| \times d_{\text{kg}}} \tag{16}$$

where $H^{(L)} = [h_1^{(L)}; \ldots; h_{|\mathcal{E}'|}^{(L)}]$ and W_g projects the graph node embeddings into the decoder feature space.

In this way, the graph-derived representation preserves entity-level relational semantics and is injected into the decoder as graph-token memory, enabling structured semantic priors to directly participate in caption generation.

E. Change-Aware Language Generator

1) Change-Conditioned Fusion and Generation: We first organize the motion-level change representation F_{disk} and the semantic-level change representation F_{sem} into an auxiliary change-token sequence, rather than collapsing them into a single global vector. After linear projection, the two types of change cues are concatenated as

$$A = \text{Concat}\left(W_d F_{\text{disk}}, W_s F_{\text{sem}}\right) \tag{17}$$

where A denotes the auxiliary change-token sequence and the graph-derived representation F_{kg} is retained as graph-token memory.

Given the global bi-temporal embedding $E_{\text{img}} \in \mathbb{R}^{h \times w \times C_e}$, we flatten it into visual token embeddings $X \in \mathbb{R}^{(hw) \times d_e}$. Instead of directly concatenating auxiliary priors with visual tokens, we first perform residual cross-attention between the auxiliary change tokens A and the visual tokens X

$$A_{i2g} = \text{RCA}\left(X,\ K = A,\ V = A\right) \tag{18}$$

$$A_{g2i} = \text{RCA}\left(A,\ K = X,\ V = X\right) \tag{19}$$

where $\text{RCA}(\cdot)$ denotes residual cross-attention. The decoder memory is then formed by concatenating the visual tokens, the bidirectionally refined auxiliary tokens, and the graph-token memory

$$M = \left[X;\ A_{i2g};\ A_{g2i};\ F_{\text{kg}}\right]. \tag{20}$$

Let $Y^{(0)} \in \mathbb{R}^{T \times d_f}$ denote the initial caption token embeddings. The decoder consists of L Transformer layers, each performing masked SA over caption tokens and cross-attention to the unified memory M

$$\widetilde{Y}^{(\ell)} = \text{SelfAttn}\left(Y^{(\ell-1)}\right) \tag{21}$$

$$Z^{(\ell)} = \text{CrossAttn}\left(\widetilde{Y}^{(\ell)},\ K{=}M,\ V{=}M\right) \tag{22}$$

$$Y^{(\ell)} = \text{FFN}\left(\text{LN}\left(\widetilde{Y}^{(\ell)} + Z^{(\ell)}\right)\right). \tag{23}$$

Finally, the hidden states $H = Y^{(L)}$ of the last decoder layer are projected onto vocabulary logits and probabilities

$$\text{logits} = W_o H + b_o, \quad P = \text{Softmax}\left(\text{logits}\right). \tag{24}$$

2) Loss Function: We train the decoder via teacher forcing with the standard cross-entropy (CE) objective against target caption tokens $\{y_t^*\}_{t=1}^T$

$$\mathcal{L}_{\text{CE}} = -\sum_{t=1}^{T} \log P\left(y_t^* \mid y_{<t}, M\right) \tag{25}$$

where M denotes the unified decoder memory composed of bi-temporal visual tokens, cross-attended auxiliary change tokens, and graph-token memory. To improve model generalization, we optionally employ label smoothing with factor ε: each ground-truth token is represented as $(1 - \varepsilon)$ for the correct word and $\varepsilon/(V - 1)$ evenly distributed across the remaining vocabulary tokens, where V denotes the vocabulary size.

IV. Experiments

A. Datasets and Evaluation Metrics

1) LEVIR-CC: The LEVIR-CC dataset [1] comprises 10 077 pairs of bi-temporal remote sensing images (5038 with changes and 5039 without changes) derived from the LEVIR-CD dataset [56]. Each image has a 256 × 256 pixel size with 0.5-m/pixel resolution, acquired from 20 regions across Texas via Google Earth API with a 5–14-year time span between acquisitions. Each image pair is annotated with five descriptive sentences (50 385 total captions), with fixed sentences for nonchange pairs and varied descriptions for change pairs. Following the default experimental settings [1], we split the dataset into 6815 pairs for training, 1333 for validation, and 1929 for testing.

2) Dubai-CC: The Dubai-CC dataset [4] consists of 500 pairs of bi-temporal remote sensing images capturing urbanization changes in the Dubai area. The images were acquired by the Enhanced Thematic Mapper Plus (ETM+) sensor onboard Landsat 7 on May 19, 2000, and June 16, 2010. The original images were cropped into 50 × 50 pixel tiles, with each pair annotated with five different change descriptions referencing Google Maps and publicly available documents, resulting in 2500 independent descriptions. Following the default experimental settings [4], we split the

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TABLE I
COMPARISONS WITH STATE-OF-THE-ART RSICC METHODS ON THE LEVIR-CC DATASET. THE BEST AND SECOND-BEST RESULTS ARE INDICATED IN BOLD AND UNDERLINED, RESPECTIVELY

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D
Capt-Rep-Diff [3]	72.90	61.98	53.62	47.41	34.47	65.64	110.57
Capt-Att [3]	77.64	67.40	59.24	53.15	36.58	69.73	121.22
Capt-Dual-Att [3]	79.51	67.23	57.46	36.56	37.16	69.19	124.42
DUDA [3]	81.44	72.22	64.67	57.79	37.15	71.04	124.32
MCCFormer-S [51]	79.90	70.26	62.68	56.36	39.60	69.46	120.39
MCCFormer-D [51]	80.42	70.87	62.86	56.38	39.91	70.44	124.44
RSICCFormer-C [1]	83.09	74.32	66.66	62.41	38.70	73.60	132.62
PSNet [52]	83.86	75.13	67.89	62.11	38.80	73.60	132.62
Chg2Cap [5]	84.43	76.35	69.12	62.98	39.42	74.34	136.25
SEN [7]	85.10	77.05	<u>70.01</u>	<u>64.09</u>	39.59	74.57	136.02
Diffusion-RSCC [53]	-	-	-	60.90	37.80	71.50	125.60
RingMoGPT [54]	83.16	74.22	66.74	60.67	**40.25**	73.97	135.32
ChangeChat [11]	83.14	-	-	-	38.73	74.01	136.56
DeltaVLM [55]	**85.78**	<u>77.15</u>	69.24	62.51	39.47	**75.01**	<u>136.72</u>
Ours	<u>85.69</u>	**77.85**	**71.03**	**65.50**	<u>39.92</u>	<u>74.77</u>	**137.50**

TABLE II
COMPARISONS WITH STATE-OF-THE-ART RSICC METHODS ON THE DUBAI-CC DATASET

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D
DUDA [3]	58.82	43.59	33.63	25.39	22.05	48.34	62.78
MCCFormer-S [51]	52.97	37.02	27.62	22.57	18.64	43.29	53.81
MCCFormer-D [51]	64.65	50.45	39.36	29.48	25.09	51.27	63.09
RSICCFormer-C [1]	67.92	53.61	41.37	31.28	25.41	51.96	66.54
Diffusion-RSCC [53]	-	-	-	33.30	27.40	56.50	88.70
SEN [7]	70.95	57.28	45.81	36.25	26.62	55.95	91.77
Sparse Focus [19]	67.30	55.97	47.00	37.30	26.32	56.38	91.59
Prompt-CC [8]	70.03	58.41	49.44	40.32	26.48	55.82	85.44
Chg2Cap [5]	72.04	60.18	50.84	41.70	28.92	58.66	92.49
Ours	**74.25**	**62.12**	**51.77**	**42.21**	**29.05**	**59.58**	**93.26**

dataset into 300 pairs for training, 50 for validation, and 150 for testing.

3) WHU-CDC: The WHU-CDC dataset [13] contains 7434 high-resolution bi-temporal image pairs spanning from 2011 to 2016, which describe changes in buildings, parking lots, roads, and other categories. In total, the dataset provides 37 170 descriptive sentences. Following the default experimental settings [13], we split the dataset into 5947 pairs for training, 743 for validation, and 744 for testing.

4) Evaluation Metrics: BLEU-N [57] measures the n-gram precision between the generated sentences and the reference sentences, emphasizing local lexical overlap. ROUGE-L [58] computes recall based on the longest common subsequence, reflecting the extent to which the generated text covers the content of the references. METEOR [59] takes both precision and recall into account, incorporating stemming, synonym matching, and a fluency penalty, thereby better capturing semantic accuracy. CIDEr-D [60], specifically designed for image captioning tasks, employs TF-IDF weighting for n-grams to highlight rare yet informative phrases.

B. Implementation Details

The deep learning methods proposed in this study are implemented using the PyTorch [63] framework, with all model training and evaluation conducted on an NVIDIA RTX 4090 GPU equipped with 24 GB of memory. During training, the Adam optimizer is employed with an initial learning rate of 0.00001, which is decayed by a factor of 0.5 after 5 epochs. The maximum number of training epochs is set to 50. After each epoch, the model is validated on the development set, and the checkpoint achieving the highest BLEU-4 score is retained as the final model for evaluation on the test set. More details can be found in our *source code*.

C. Comparison on Public Benchmark Datasets

As shown in Tables I–III, we conducted comparisons with several existing state-of-the-art methods on three key RS-CC datasets (LEVIR-CC, Dubai-CC, and WHU-CDC). The performance of our method, SAGE-CC, was evaluated using multiple evaluation metrics, including BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, ROUGE-L, and CIDEr.

On the LEVIR-CC dataset, SAGE-CC ranks among the top performers across multiple metrics, with BLEU-2 at 77.85,

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TABLE III
COMPARISONS WITH STATE-OF-THE-ART RSICC METHODS ON THE WHU-CDC DATASET

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D
DUDA [3]	79.04	69.53	61.57	55.64	34.29	68.98	121.85
MCCFormer-S [51]	82.14	76.29	71.08	66.51	43.50	79.76	148.88
MCCFormer-D [51]	73.29	67.88	64.03	60.96	39.69	73.67	134.92
RSICCFormer-C [1]	78.25	72.82	68.57	65.14	44.35	76.50	143.44
MaskApproxNet [20]	81.34	75.68	71.16	67.73	43.89	75.41	135.31
CTMTNet [13]	83.56	77.66	72.76	69.00	45.39	79.23	149.40
Semantic-CC [10]	82.77	76.32	71.59	68.43	44.49	78.23	150.23
KCFI [12]	83.34	77.27	72.40	68.47	44.95	79.59	149.32
PTNet [61]	83.94	77.89	72.94	69.37	45.69	79.64	150.02
CTM [62]	85.36	79.49	75.36	72.36	46.98	**80.97**	153.29
Ours	**86.04**	**81.16**	**77.28**	**74.42**	**47.89**	80.90	**156.21**

TABLE IV
ABLATION STUDY ON THE LEVIR-CC DATASET, ASSESSING THE IMPACT OF KEY COMPONENTS: MCL, SG, SCA, AND REMOTE SENSING CGR. SYMBOLS "×" AND "✓" INDICATE THE EXCLUSION AND INCLUSION OF SPECIFIC MODULES, RESPECTIVELY. HIGHER METRIC VALUES DENOTE BETTER PERFORMANCE, WITH BEST RESULTS HIGHLIGHTED IN BOLD

Method	MCL	SG	SCA	CGR	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D
Baseline	×	×	×	×	84.43	76.35	69.12	62.98	39.42	74.34	136.25
(a)	✓	×	×	×	85.05	76.59	69.50	63.66	39.71	74.63	137.01
(b)	✓	✓	×	×	85.22	76.74	70.09	64.22	39.65	74.20	137.31
(c)	✓	✓	✓	×	**85.82**	77.66	70.46	64.47	39.83	74.66	**138.01**
(d)	✓	✓	✓	✓	85.69	**77.85**	**71.03**	**65.50**	**39.92**	**74.77**	137.50

BLEU-3 at 71.03, and BLEU-4 at 65.50. Additionally, the CIDEr score reaches 137.50, demonstrating the effectiveness of our method in generating descriptions highly consistent with human annotations. On the Dubai-CC dataset, SAGE-CC also exhibits strong performance, achieving excellent results across all evaluation metrics. The BLEU-1 score is 74.25, and the BLEU-4 score is 42.21. Furthermore, the CIDEr score of 93.26 further highlights the advantage of our model in generating high-quality descriptions. On the WHU-CDC dataset, SAGE-CC again leads with a BLEU-1 score of 86.04 and a BLEU-4 score of 74.42, with a CIDEr score as high as 156.21. These results further validate the superiority of our method in generating precise and detailed remote sensing image descriptions.

D. Ablation Study

As shown in Table IV, we systematically evaluated the contributions of four key components to the overall performance of the model on the LEVIR-CC dataset, namely MCL (without SG), SG, semantic-level change aggregation (SCA), and the remote sensing CGR. The results indicate that, compared with the baseline model (Baseline), the overall model performance shows an improving trend as these modules are progressively added, although the gains vary across different NLG metrics (BLEU-1 to BLEU-4, METEOR, ROUGE-L, and CIDEr-D). This demonstrates that each component enhances the model's ability to capture and describe changed regions from different perspectives.

Specifically, after introducing MCL, Method (a) improves BLEU-4 from 62.98 to 63.66 and CIDEr-D from 136.25 to 137.01, indicating that motion-level changed regions help generate more accurate change descriptions. After integrating SG, Method (b) achieves improvements in BLEU-1, BLEU-2, BLEU-3, BLEU-4, and CIDEr-D, while METEOR and ROUGE-L decrease slightly. This suggests that SG can enhance local feature matching and fine-grained correspondence modeling between bi-temporal images, thereby improving most generation metrics, although its effect is not reflected consistently across all evaluation metrics. With the further addition of SCA, Method (c) increases BLEU-4 from 64.22 to 64.47, while METEOR and ROUGE-L also recover to some extent, indicating that semantic-level assistance plays a positive role in change representation and caption generation. Finally, Method (d), which incorporates the CGR module, further boosts BLEU-4 to 65.50 and achieves the best overall performance on most metrics, demonstrating the robustness and effectiveness of the overall framework. Meanwhile, the slight fluctuations across some metrics indicate that RS-CC is inherently a multiobjective task, where improvements in fine-grained matching, semantic completeness, and structural consistency may not always be fully synchronized.

In summary, the experimental results verify the effectiveness of all the proposed modules in improving model performance. In particular, the combination of MCL and SG effectively strengthens fine-grained change modeling and brings gains on most major generation metrics. However, relying solely on enhanced feature matching cannot guarantee simultaneous improvements in semantic completeness and long-sequence structural consistency. SCA provides valuable support at the semantic level by aggregating higher-level semantic change information, which further alleviates the potential semantic insufficiency of local matching results and thus helps improve the completeness and rationality of the generated descriptions.

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TABLE V
ABLATION STUDY ON THE NUMBER OF MASKS IN THE MCL MODULE ON THE LEVIR-CC DATASET

Number	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr
30	85.13	76.98	69.83	63.74	39.81	74.60	137.09
40	85.31	77.07	70.15	**64.48**	39.52	74.32	136.63
50	**85.82**	**77.66**	**70.46**	64.47	**39.83**	**74.66**	**138.01**

TABLE VI
ABLATION STUDY ON THE THRESHOLD OF SG ON THE LEVIR-CC DATASET

Threshold	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr
0.1	84.82	76.70	69.57	63.57	39.71	74.35	137.13
0.2	**85.82**	**77.66**	**70.46**	**64.47**	**39.83**	**74.66**	**138.01**
0.3	85.03	76.94	69.73	63.87	39.48	73.88	134.39

TABLE VII
ABLATION STUDY ON DIFFERENT GRAPH ENCODERS FOR MODEL PERFORMANCE ON THE LEVIR-CC DATASET

Encoder	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr
GCN	85.23	77.01	69.73	63.80	39.56	74.17	135.62
GAT	85.42	77.20	70.21	64.51	39.59	74.37	136.20
SAGE	84.06	75.91	68.87	63.10	39.26	73.79	134.80
MONET	82.71	74.15	67.16	61.64	38.22	72.58	130.20
RGCN	**85.69**	**77.85**	**71.03**	**65.50**	**39.92**	**74.77**	**137.50**

In contrast, the CGR module, through structured information extraction and reasoning, further models changed objects and their relationships, providing clearer semantic constraints and contextual support for the generation process, thereby improving the accuracy and consistency of remote sensing change descriptions.

E. Component Analysis

1) Analysis of Numbers of Masks in the MCL Module: As shown in Table V, we investigated the impact of different numbers of masks in the MCL module on the model's performance. The results indicate that as the number of masks increases from 30 to 50, the overall performance of the model gradually improves. Specifically, when the number of masks is 50, the model achieves optimal performance in metrics such as BLEU-1, BLEU-2, BLEU-3, and CIDEr, with BLEU-4 and CIDEr reaching 64.47 and 138.01, respectively. This suggests that increasing the number of masks appropriately allows for a more comprehensive coverage of motion-level changed regions, thereby providing richer fine-grained features for change detection. When the number of masks is too small, the model struggles to capture sufficient change information, leading to a decrease in generation performance.

2) Analysis of Matching Threshold in the SG Module: As shown in Table VI, we further explored the impact of different matching threshold settings in the SG module on the model's performance. The experimental results demonstrate that when the threshold is set to 0.2, the model reaches a peak CIDEr score of 138.01, while BLEU-4 and ROUGE-L also maintain their highest levels. A threshold that is too low (0.1) or too high (0.3) leads to a performance decline. The former may introduce excessive irrelevant information, while the latter may overlook valid correspondences. Therefore, setting an appropriate threshold is crucial for balancing robustness and precision.

3) Comparison of Graph Encoding Methods: As shown in Table VII, we compared the performance of different graph encoding methods (GCN, GAT, SAGE, MONET, and RGCN) on BLEU, ROUGE-L, METEOR, and CIDEr metrics. The results show that the RGCN achieves the best performance across all metrics, with BLEU-4, ROUGE-L, and CIDEr reaching 65.50, 74.77, and 137.50, respectively, which are significantly superior to other methods. In contrast, methods such as the GCN exhibit relatively weaker performance, indicating their limitations in handling complex relational dependencies. We speculate that the RGCN can explicitly model multiple relation types, thereby better capturing the dependency structures between different change regions and semantic units. This plays a crucial role in generating more coherent and accurate natural language descriptions.

4) Analysis of Threshold k for Filtering Low-Frequency Relations and Entities: As shown in Table VIII, we further investigated the impact of the threshold k for filtering low-frequency relations and entities on the model's performance. The experimental results demonstrate that as k increases from 30 to 50, the model's performance generally improves. When $k = 50$, the model achieves the best performance in BLEU, with BLEU-1, BLEU-2, BLEU-3, and BLEU-4 reaching 85.69, 77.85, 71.03, and 65.50, respectively, significantly outperforming other settings. When k is further increased to 60, BLEU performance shows a slight decline, but ROUGE-L,

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TABLE VIII
ABLATION STUDY ON THE FREQUENCY THRESHOLD k FOR FILTERING LOW-FREQUENCY RELATIONSHIPS AND ENTITIES ON THE LEVIR-CC DATASET

k	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr
30	84.21	76.15	69.26	63.50	39.23	73.83	134.90
40	85.10	77.25	70.29	64.58	39.82	74.55	135.49
50	**85.69**	**77.85**	**71.03**	**65.50**	39.92	74.77	137.50
60	85.22	77.41	70.45	64.63	**40.28**	**75.23**	**138.21**
70	85.18	77.06	69.92	64.14	39.78	74.39	135.69

TABLE IX
PARAMETER COMPARISON AND PERFORMANCE ON THE LEVIR-CC DATASET

Method	Params(M)	BLEU-4	CIDEr-D
MCCFormers-S	69.86	59.07	122.06
RSICCFormer-C	172.80	62.41	132.62
PSNet	424.39	62.11	132.62
RSCaMa	790.18	65.24	136.56
Prompt-CC	408.58	63.54	136.44
Sparse Focus	687.10	62.87	137.05
Ours	385.80	65.50	137.50

METEOR, and CIDEr improve to 75.23, 40.28, and 138.21, respectively. As k continues to increase, all performance metrics decline, suggesting that an excessively high threshold may overly filter out some valid relations and entities, thus weakening the model's expressive ability. In summary, $k = 50$ is the optimal setting in this experiment, effectively balancing the retention of semantic richness while filtering out low-frequency irrelevant information.

F. Efficiency Analysis

As shown in Table IX, SAGE-CC is not the most lightweight method on the LEVIR-CC dataset in terms of parameter scale. In our implementation, SAGE-CC contains 385.80 M parameters, with 50.6 GFLOPs and an inference latency of 136.6 ms. For example, MCCFormers-S and RSICCFormer-C contain fewer parameters than our method. This is mainly because SAGE-CC introduces additional region-aware and knowledge-guided components to model fine-grained changed regions, semantic cues, and structured change priors. Nevertheless, compared with several recent competitive methods, such as PSNet, RSCaMa, Prompt-CC, and Sparse Focus, the parameter scale of SAGE-CC remains relatively moderate. More importantly, SAGE-CC achieves competitive BLEU-4 and CIDEr-D scores, indicating that the increased model capacity is effectively translated into improved CC performance.

These results demonstrate that the proposed region-aware change mining module and graph-guided reasoning module contribute positively to semantic change description, rather than merely increasing the model size. Therefore, SAGE-CC provides a reasonable tradeoff between model complexity and captioning performance, achieving stronger CC accuracy with an acceptable parameter scale.

G. Visualization

1) Captioning Results: As shown in Fig. 3, we present some examples to demonstrate the effectiveness of our proposed SAGE-CC model in RS-CC. For specific dual-temporal remote sensing images, we compare the captions generated by the SAGE-CC model with those generated by the baseline model and the ground truth labels. To provide a more intuitive visualization, we use red font to indicate the inaccurate descriptions generated by the baseline model, while green font represents the portions of the captions generated by the SAGE-CC model that are consistent with the ground truth. From the visualization results, it is evident that the captions generated by our model are of higher quality and more consistent with the ground truth compared to those generated by the baseline model.

2) Visualization of Semantic Segmentation Masks: As shown in Fig. 4, we present the semantic segmentation results obtained using the SAM on dual-temporal remote sensing images. Since there are many semantic categories but limited display space, we select the three most frequently appearing classes for visualization. In the segmentation maps, blue regions represent buildings, green regions indicate vegetation, and pink regions correspond to roads. By applying SAM-based segmentation to both temporal images, fine-grained object boundaries and class-level changes can be effectively captured, providing rich spatial and structural information for subsequent CC tasks.

3) KG Constructed From Captions: As shown in Fig. 5, due to the large number of nodes, we only display a portion of the relationships between the nodes to ensure the clarity and readability of the graph. Each node in the figure represents an entity, and each arrow indicates a relationship between entities, with different arrow colors representing different types of relationships. This approach allows for a more intuitive visualization of the key information and interrelationships in remote sensing captions, helping to understand the complex patterns and potential connections within the captions.

4) Visualization of SG: As shown in Fig. 6, we present some matching results generated by the SG algorithm. The red points represent the mask feature points extracted from the two images, while the green lines indicate the successfully matched feature point pairs, demonstrating the spatial correspondence between the two images. These matches provide crucial support for subsequent change detection tasks.

H. Limitation Analysis

Although SAGE-CC benefits from SAM-guided region mining and graph-guided semantic reasoning, these modules do not fully resolve all challenges in RS-CC. As shown in Fig. 7, when changed objects are occluded by shadows, the SAM may fail to clearly segment the affected regions. Similarly, when changed objects appear near image boundaries, the semantic

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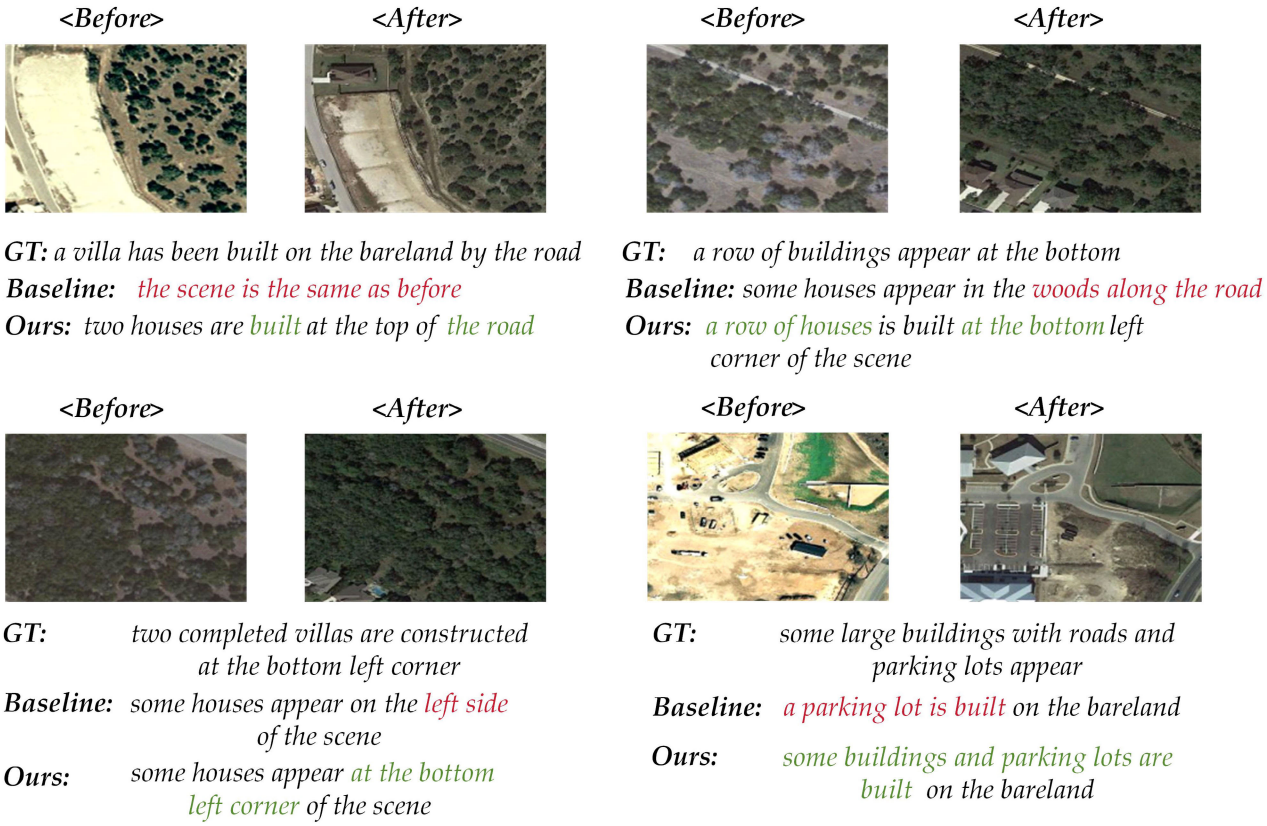


Fig. 3. Remote sensing change image pairs and their corresponding ground-truth annotations are provided in the LEVIR-CC dataset. The matching parts are shown in green, and the mismatched parts are shown in red.

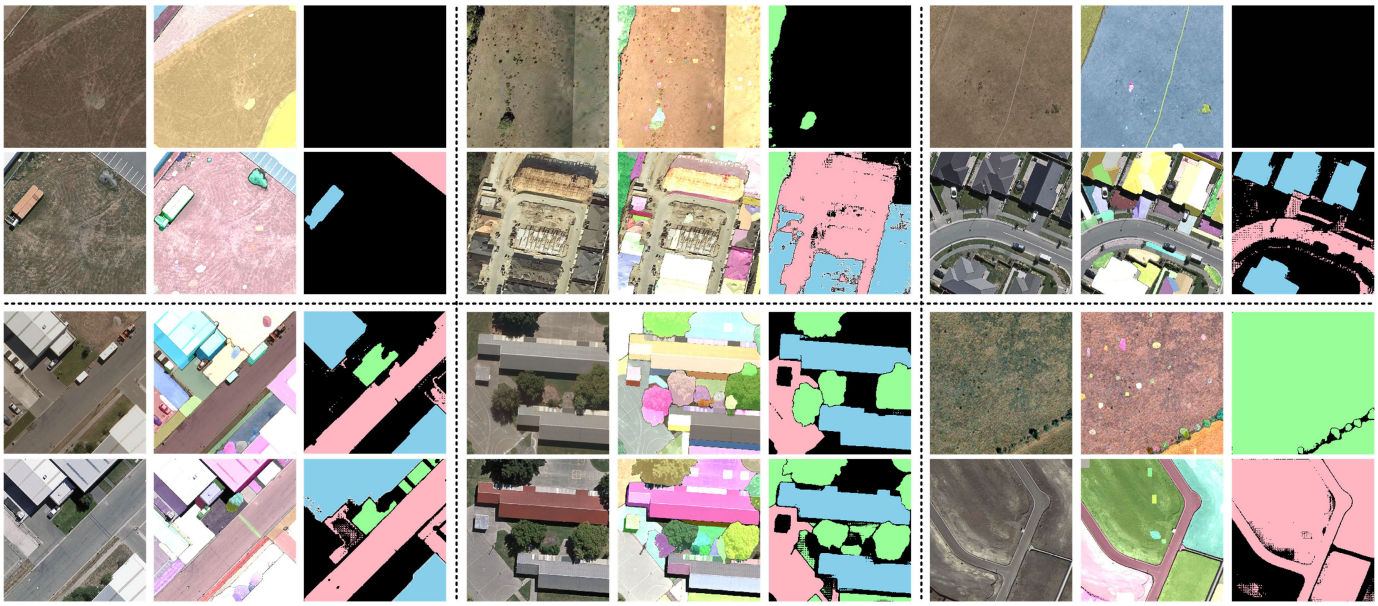


Fig. 4. Semantic segmentation results obtained using the SAM model. Blue, green, and pink regions denote building, vegetation, and road, respectively.

segmentation results may be incomplete or partially missing. These issues can further hinder accurate change modeling and caption generation. In addition, in complex scenes, especially when encountering rare change patterns or subtle semantic differences, the semantic constraints provided by the graph prior could be further enriched. Although SAM-guided region mining, SG-based matching, and graph reasoning are jointly organized through entity- and relation-level correspondences, the integration between visual change cues and text-based graph semantics can still be further strengthened, particularly in terms of structural hierarchy and semantic representation. Meanwhile, since the overall framework contains multiple

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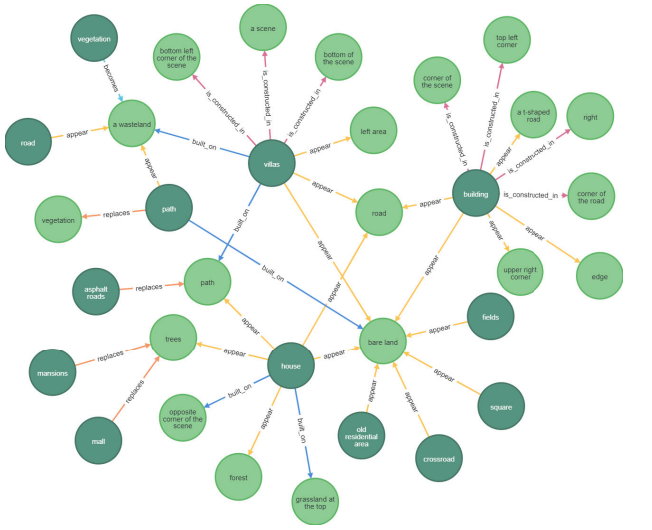


Fig. 5. Illustration of the built KG for the RS-CC task. Zoomed-in view for better visualization.

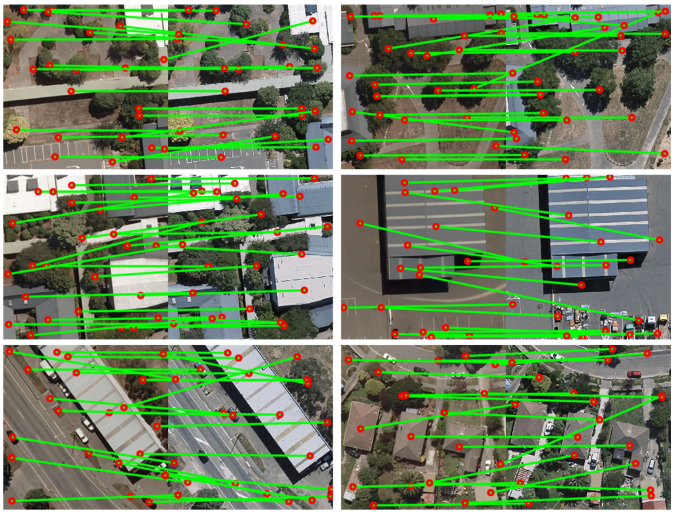


Fig. 6. Visualization of the matching results by SG.

modules, the computational cost is relatively high, which may limit its deployment and application in resource-constrained scenarios.

V. Conclusion

In this article, we propose a novel approach to RS-CC that leverages the SAM to extract region-level representations and integrates them with auxiliary knowledge sources to enhance change description. By fusing visual and semantic information through a Transformer architecture, we generate natural language descriptions with high geometric accuracy and semantic richness. Extensive experimental results demonstrate that our method achieves competitive performance across multiple public benchmark datasets and provides effective improvements on several important evaluation metrics.

Acknowledgment

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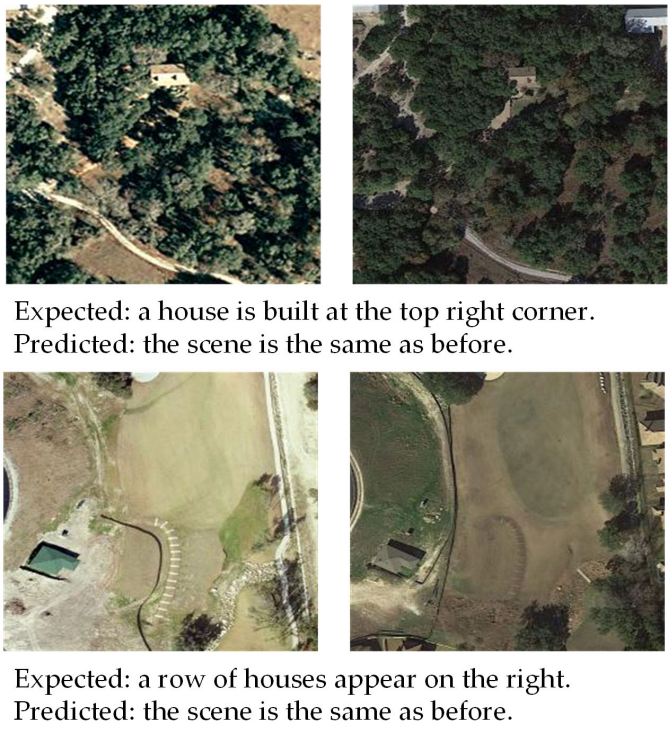


Fig. 7. Visualization of representative failure cases.

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